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Open Ended LAB

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Subject: Data Structures Lab

Objective:

The objective of this lab task is to design and implement a basic hospital management system that:

- Adds patients with department and type categorization.
- Assigns a priority to each patient based on urgency.
- Retrieves and treats the most critical patient first.
- Allows patient information to be updated (e.g., transferring departments).
- Displays the list of patients in the queue.

Theory:

In real-world hospital environments, patients are treated not on a strict first-come-first-served basis but based on the severity or urgency of their condition. This program simulates that concept using a priority system implemented via structured programming in C++.

Key Concepts:

1. **Structures and Classes:**
   - struct Patient holds details such as ID, name, department, type, and computed priority.
   - class Department encapsulates all functionalities for managing patients.
2. **Priority Assignment:**
   - Departments and patient types influence priority:
     - *Emergency* patients get a base score of 100.
     - *General* and *Pediatrics* get 50.
     - *VIP*, *Elderly*, and *Normal* types add 30, 20, and 10 respectively.
   - Total priority = Department Score + Type Score.
3. **Operations:**
   - **Add Patient:** Takes input, assigns priority, and stores the patient.
   - **Call Next Patient:** Searches for the highest-priority patient and removes them from the queue.
   - **Display Queue:** Shows all current patients with full details.
   - **Transfer Patient:** Updates a patient's department and reassigns priority.
4. **Control Flow:**
   - A simple menu-driven interface within the main() function allows interaction with all system functionalities.

Conclusion:

This open-ended lab task successfully demonstrated how basic object-oriented programming and conditional logic can be used to simulate real-life scenarios such as patient triage in hospitals. The

implementation allowed dynamic patient addition, priority-based calling, updating, and viewing—mimicking a real hospital's decision-making system. The program improves understanding of:

- Structure and class usage in C++
- User input and validation
- Priority logic
- Array manipulation and linear search

Overall, the task highlights how computational logic can be effectively used in healthcare management systems.